One Head Away from Language Confusion: Single Attention Heads Across Model Families

Anonymous ACL submission

Abstract

Large language models sometimes reply in a language other than the one the user wrote in or requested, a failure known as language confusion. We ablate individual attention heads in 12 instruction-tuned models (1B–7B parameters), using FLORES-200 continuations in five languages and Language Confusion Benchmark (LCB) chat replies. In ten models, zeroing one head causes substantially more departures from the expected language than zeroing any other head in its layer; its LCB pass-rate drop persists, though reduced, under mean ablation. Seven such heads move 8.0–51.5% of FLORES continuations out of the prompt's language, versus at most 1.4% for other same-layer heads; the other three mainly affect explicitly requested languages. These seven heads have smaller effects in base models given plain text. In two models, replacing the values a head reads at the requested-language name with another language's mean values makes replies fail and shifts some into that language, unlike a random change of equal size. Replacing a head's output with another language's mean redirects 84–99% of non-English monolingual LCB replies into the replacement language in five models. Yet the requested language's mean improves crosslingual replies beyond every control head only in the two Gemma models, whose baselines are low.

1 Introduction

Large language models (LLMs) are expected to reply in the language of the prompt or in a language explicitly requested by the user, yet they sometimes reply in another language. Marchisio et al. (2024) call this failure *language confusion* and measure it with the Language Confusion Benchmark (LCB). In their evaluation, base and English-centric instruction-tuned models are more prone to it, and even the strongest models do not always reply in the expected language. What happens inside the model when a reply leaves the expected language is less well understood.

Mechanistic studies suggest that output-language information is partly separable from content and that later layers contribute to language-specific decoding (Wendler et al., 2024; Dumas et al., 2025; Harrasse et al., 2025). Prior work has also shown that output language can be steered or repaired through small sets of neurons (Tang et al., 2024; Nie et al., 2025) or attention heads (Liu et al., 2026), including heads involved in target-language identification during few-shot translation in base models (Lasnier et al., 2026b). It remains unclear whether language adherence in instruction-tuned chat replies can be disproportionately sensitive to a single head, whether similar single-head dependencies appear across model families, and whether heads found on text continuations also affect the language of chat replies. A large language-adherence effect need not coincide with a large increase in language-modeling loss: zeroing Qwen2.5-7B's L19H1 raises the loss by only 0.026 nats but moves 20% of FLORES continuations out of the prompt's language.

We address these questions by ablating attention heads one at a time in 12 instruction-tuned models from seven model series (1B–7B parameters), using FLORES-200 continuations in five languages and monolingual and crosslingual chat prompts from LCB (Figure 1). We distinguish correct-to-wrong from wrong-to-correct language changes, set aside heads whose removal sharply raises language-modeling loss or frequently moves English continuations out of English, and compare each selected head with the other heads in its layer. For models without a qualifying head on FLORES, we screen all heads on LCB. We then compare the seven FLORES heads with their base checkpoints, examine what two heads read, and replace head outputs with language-conditioned means. Our main findings are:

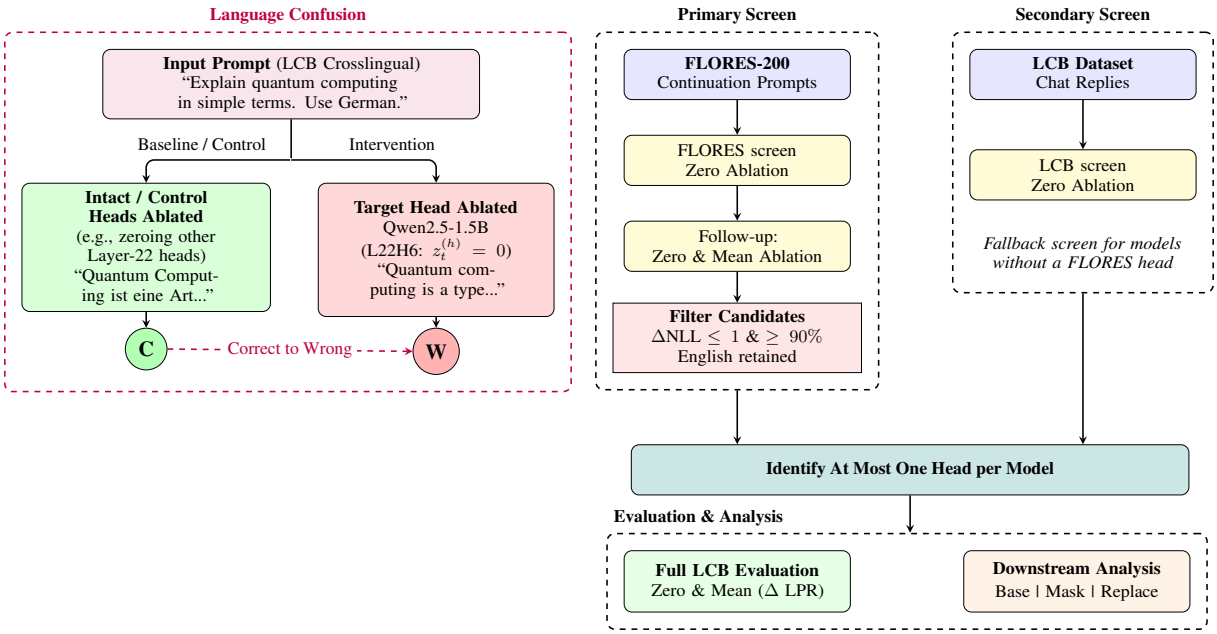


Figure 1: Methodology for identifying and analyzing at most one head per instruction-tuned model, accompanied by a real-world demonstration of the correct-to-wrong language confusion metric.

(1) In ten of the twelve models, zeroing one head causes far more departures from the expected language than zeroing any other head in its layer. Its effect on LCB language adherence persists, though reduced, under mean ablation. Seven of these heads move 8.0–51.5% of FLORES continuations out of the prompt's language; the other three mainly affect explicitly requested languages (Section 4).

(2) The seven FLORES heads have smaller effects in the base models given plain text; in Qwen2.5-1.5B, the difference appears mainly with the chat template (Section 5).

(3) In Qwen2.5-1.5B and Gemma-3-1B, replacing the values a head reads at tokens naming the requested language with mean values from prompts naming another language makes replies fail and shifts some into that language, which a random change of the same size does not do. Masking the head's attention to these tokens also reproduces part of the effect of removing the head (Section 6.1).

(4) Replacing a head's output with its mean for another language puts 84–99% of scored replies to non-English monolingual LCB prompts entirely in the replacement language in five models. However, the requested language's mean improves crosslingual replies beyond every control head only in the two Gemma models (Section 6.2).

2 Related Work

Head ablation. Mechanistic analyses study how internal components contribute to model behavior (Elhage et al., 2021) and localize some behaviors in small circuits of attention heads (Wang et al., 2022; Conmy et al., 2023). Many heads can be removed with little performance loss, sometimes reducing layers to a single head (Michel et al., 2019), while specialized heads in machine translation tend to survive pruning (Voita et al., 2019). Some capabilities, however, depend on small subsets of heads: retrieval heads constitute fewer than 5% of the heads in the models studied, yet their ablation substantially impairs long-context retrieval, unlike ablating random non-retrieval heads (Wu et al., 2024). We investigate the effects of single-head ablation on output-language behavior rather than aggregate task performance.

Multilingual representations. In Llama-2, analyses of intermediate layers suggest a concept space closer to English than to other languages (Wendler et al., 2024), while activation patching shows that the output language is encoded at an earlier layer than the concept (Dumas et al., 2025). Language-specific decoding emerges in later layers (Harrasse et al., 2025), and cross-lingual factual inconsistencies have been linked to failures in the late transition to language-specific representations (Wang et al., 2025). Key language-related neurons also

concentrate in the first and last layers (Zeng et al., 2025). These findings motivate screening attention heads across all layers rather than at a preselected depth.

Language-selective neurons. Tang et al. (2024) identify language-specific neurons, mostly in the first and last layers, and steer output languages by activating or deactivating them. Gurgurov et al. (2025) study the identification and manipulation of such neurons across 21 languages. Tezuka and Inoue (2025) propose that specific MLP neurons mediate transitions between language-specific latent spaces and a shared semantic space. Among neuron-level studies, Nie et al. (2025) trace language confusion in English-centric models to transition failures in the final layers and mitigate them by editing neurons identified through comparison with a multilingual-tuned counterpart. We examine whether individual attention heads can have similarly large causal effects on language adherence in instruction-tuned models.

Attention heads and language control. LAHIS (Liu et al., 2026) identifies language-specific and language-general heads in Aya-23-8B, Llama-3.2-3B, and Mistral-7B-v0.1, using language-specific heads to reduce off-target generation. Lasnier et al. (2026b) study few-shot translation in base models without explicit instructions. Through activation patching, they identify distinct sparse head sets for target-language identification and meaning preservation. Steering these heads enables instruction-free translation, while their ablation selectively disrupts the corresponding functions.

Their language heads provide a close precedent for ours. In their released base-model scores (Lasnier et al., 2026a), averaged over 20 translation directions, the heads we identify in instruction-tuned Qwen3-1.7B and Llama-3.2-1B each rank first within their models, while the Gemma-3-1B head ranks third (Appendix H). Because the scores come from base models, this agreement does not establish identical functions.

Building on these studies, we examine whether keeping chat replies in the expected language is disproportionately sensitive to individual heads in instruction-tuned models, both when that language is implicit and when it is explicitly requested. We further compare these dependencies with base checkpoints, investigate what selected heads read, and test whether replacing their outputs can restore the requested language.

Language confusion and inference-time steering. LCB (Marchisio et al., 2024) evaluates language confusion in monolingual and crosslingual prompts, including explicit requests for a reply language; we use its prompts and line-level pass rate. Oh et al. (2026) find failures in inferring the expected reply language from code-switched prompts, including biases toward non-English replies and within-response language switching. Language-specific steering vectors extracted from a multi-parallel corpus can reduce language confusion while preserving task performance (Sterz et al., 2025). Function vectors add task-conditioned mean outputs of several heads to the residual stream to induce a task (Todd et al., 2024). We instead replace one head's output with its mean for a language to probe its role in output-language control, rather than to propose a general mitigation method.

3 Setup

3.1 Models and Data

Our main evaluation covers 12 instruction-tuned models across seven model series (1B–7B parameters): Qwen2.5 1.5B, 3B and 7B (Yang et al., 2024), Qwen3 1.7B and 4B (Yang et al., 2025), Gemma 3 1B and 4B (Gemma Team, 2025), OLMo 2 1B (Team OLMo et al., 2024), OLMo 3 7B (Team Olmo et al., 2025), Llama 3.2 1B and 3B (Grattafiori et al., 2024), and SmolLM3 3B (Bakouch et al., 2025).

For implicit continuation, we use 2,500 prompts from FLORES-200 (NLLB Team et al., 2022), 500 each in English, French, German, Spanish and Italian: the first 495 aligned devtest sentences that pass a length and letter-ratio filter, and five hand-written sentence starts per language (Appendix A). The first 25 prompts per language (125 in total) form the screening set.

For chat replies, we use the Language Confusion Benchmark (LCB; Marchisio et al., 2024). The five-language evaluation includes monolingual prompts written in French, German, Spanish, or Italian (800 prompts), plus 200 English prompts as controls. Crosslingual prompts are written in English and request a reply in one of the four non-English languages (1,196 prompts). Instruction-tuned models receive every prompt as one user turn in their own chat template; base checkpoints receive FLORES prompts as plain text (templates, precision and decoding in Appendix A).

3.2 Interventions

Let $z_t^{(h)}$ be the output slice of attention head h at sequence position t, immediately before the attention output projection o_{proj}, which mixes the heads' contributions across all hidden dimensions. Our primary intervention sets this slice to zero at every position, while the prompt is processed and while the reply is generated; Section 6.1 also intervenes in one phase at a time.

Since interpretability results can depend on how an activation is replaced (Zhang and Nanda, 2024), we also report mean ablation, which replaces $z_t^{(h)}$ with its mean μ_h in the unmodified model over the prompt positions of the 2,500 FLORES prompts, chat-template tokens included; the same mean is used for FLORES and LCB. For language-specific replacement, we instead average the head's output over the FLORES prompts in language ℓ and the unmodified model's greedy continuations of them, from the first token of the user's text to the end of the continuation, giving $\mu_{h,\ell}$; Section 6.2 replaces the head's output with the mean for a chosen language. Unlike μ_h, this mean also covers generated tokens; the prompt language's own mean, computed in the same way, serves as its control (Section 6.2, Appendix G).

3.3 Measuring Language Confusion

On FLORES, each model greedily generates up to 40 new tokens, and `langdetect` (Nakatani, 2010) assigns the language $\mathcal{L}(y)$ of each continuation y. Let ℓ_j be the language of prompt j, and let y_j and y_j^{-h} be the baseline and head-ablated continuations. The directional correct-to-wrong rate is

$$C \to W(h) = \frac{1}{n} \sum_{j=1}^{n} \mathbf{1}\big[\, \mathcal{L}(y_j) = \ell_j \wedge \mathcal{L}(y_j^{-h}) \neq \ell_j \big], \tag{1}$$

where n is the number of prompts in all five languages (2,500, or 125 in the screen). Unlike a raw switch rate, $C \to W$ ignores changes away from an already wrong label; the reverse change, $W \to C$, is at most 0.002 for each of the seven heads found on FLORES.

On LCB, replies are generated greedily for up to 100 new tokens and scored with the benchmark's rules (Joulin et al., 2016; Marchisio et al., 2024): fastText labels each line of at least five words, a reply passes only when all its scored lines are in the expected language, and replies with no such line are not scored. The line-level pass rate (LPR) is the pass rate averaged over the benchmark's sources. For each head, ΔLPR is the paired mean difference in pass outcomes between ablated and unablated replies over the prompts scored in both (Appendix A), computed separately for monolingual and crosslingual prompts; since it weights prompts equally, it can differ from the difference of the source-averaged LPRs. A negative value means less adherence to the expected language. Its 95% confidence interval comes from 2,000 paired bootstrap resamples of the prompts.

3.4 Choosing the Head

All 12 models are screened on FLORES; eight have a head that qualifies there and go to a 2,500-prompt follow-up, and the other four are screened on LCB (Figure 1). On the 125 screening prompts we ablate every head and apply two conditions: ΔNLL ≤ 1, where ΔNLL is the change in mean per-token loss (in nats) on 100 FLORES dev sentences per language, given as plain text and averaged over the five languages, and at least 90% of English continuations kept in English, which exclude heads whose removal disrupts generation in general or on English prompts. A model goes to the follow-up when a head that meets both conditions reaches a $C \to W$ of 0.10 on the screen. The follow-up reruns all heads of the layers that hold the strongest screened heads (Appendix B), and we report the one with the largest $C \to W$ on all 2,500 prompts among those that meet both conditions, with English retention now measured on all 500 English prompts. We compare each selected head with the other heads of its layer rather than across architectures, and in each full LCB evaluation we also ablate three randomly chosen heads of the same layer as controls (one for Llama-3.2-1B). Because the head is chosen on all 2,500 prompts, its FLORES effect is conditioned on that choice. The head itself is chosen without LCB data, so LCB tests it on a separate task. LCB did, however, decide which FLORES heads are counted: OLMo-3-7B also qualifies on FLORES, but we report it separately, by a criterion set after its LCB results that all ten counted heads meet (Section 4.1).

The LCB screen ablates every head on 100 crosslingual LCB prompts (25 per requested language). Among the heads that meet both conditions on the FLORES screen, the one with the largest drop in the pooled pass rate goes to the full LCB

Model	H	Head	FLORES $C \to W$ Zero	Mean	Others	Base	ΔNLL	LCB monolingual LPR	ΔLPR	LCB crosslingual LPR	ΔLPR
Heads found on FLORES											
Qwen2.5-1.5B	12	L22H6	0.500	0.450	0.007	0.150	+0.228	0.982	−0.273	0.704	−0.276
Qwen2.5-3B	16	L27H13	0.515	0.338	0.014	0.202	+0.234	0.982	−0.537	0.888	−0.449
Qwen2.5-7B	28	L19H1	0.200	0.018	0.006	0.025	+0.026	0.984	−0.009	0.950	−0.074
Qwen3-1.7B	16	L18H12	0.324	0.028	0.002	0.067	+0.128	0.985	−0.149	0.823	−0.508
Gemma-3-1B	4	L11H3	0.413	0.089	0.002	0.001	+0.720	0.984	−0.628	0.118	−0.091
Gemma-3-4B	8	L24H0	0.214	0.012	0.000	0.148	+0.202	0.990	−0.259	0.133	−0.107
OLMo-2-1B	16	L12H8	0.080	0.056	0.005	0.005	+0.136	0.986	−0.272	0.931	−0.335
Heads found on LCB											
Qwen3-4B	32	L24H27	0.016^s	–	0.008^s	–	+0.065	0.982	+0.000	0.893	−0.076
Llama-3.2-1B	32	L8H25	0.048^s	–	0.016^s	–	+0.067	0.997	−0.016	0.874	−0.711
Llama-3.2-3B	24	L13H19	0.000^s	–	0.000^s	–	+0.008	0.993	+0.000	0.911	−0.335
Not counted (Section 4.1)											
OLMo-3-7B	32	L14H25	0.134	0.007	0.010	0.104^s	+0.202	0.972	−0.041	0.874	−0.018
SmolLM3-3B	16	none in either screen (L1H12 breaks generation)									

Table 1: Head-ablation effects across 12 instruction-tuned models. H: heads per layer. FLORES: Zero and Mean are $C \to W$ (Eq. (1)) under zero and mean ablation; Others is the largest same-layer Zero, and Base is Zero in the base checkpoint. ΔNLL: loss change under zero ablation. LCB: baseline pass rate (LPR) on the non-English prompts and its paired change under zero ablation (ΔLPR; Section 3.3). ^{s}: 125-prompt FLORES screen; gray: 95% CI includes zero; –: not evaluated. Heads found on LCB are compared on LCB (Section 4.2); all screens are in Appendix B.

evaluation, provided the drop is at least 0.10; for these heads, the comparison with the other heads of the layer comes from this screen, and Section 4.2 reports their effect on the crosslingual prompts not used to select them. Section 4.3 reruns this screen on the models with a FLORES follow-up as a check.

4 A Single Head Stands Out in Ten Models

We count ten heads (Table 1): seven found on FLORES and three found on LCB that act mainly on explicitly requested languages. We report OLMo-3-7B's FLORES head separately, and SmolLM3-3B has no qualifying head.

4.1 Passage Continuations (FLORES)

In seven models, zeroing one head moves 8.0–51.5% of all FLORES continuations out of the prompt's language, versus at most 1.4% for any other head in its layer. Most departures go to English in Qwen2.5-1.5B, Qwen3-1.7B and Gemma-3-1B, whereas 441 of the 500 departures in Qwen2.5-7B go to Chinese (Appendix D).

We report OLMo-3-7B's L14H25 (13.4%) separately: its LCB effects are small (−0.041 and −0.018) and vanish under mean ablation (+0.001 and +0.015), whereas each of the ten counted heads still lowers at least one LCB pass rate under mean ablation (Section 4.2).

4.2 Chat Replies (LCB)

Each of the ten heads lowers at least one LCB pass rate, with the 95% CI of its paired change below zero (Table 1, Appendix C); the upper end of that interval is also below the change of every randomly chosen same-layer control head. For the seven FLORES heads, the effects are roughly balanced across monolingual and crosslingual prompts in Qwen2.5-1.5B, Qwen2.5-3B and OLMo-2-1B, mostly crosslingual in Qwen3-1.7B and Qwen2.5-7B, and mostly monolingual in the two Gemma models, whose crosslingual baselines are low. In three of the four models screened on LCB, the screen selects L8H25 (Llama-3.2-1B), L13H19 (Llama-3.2-3B) and L24H27 (Qwen3-4B). Their ablation turns 71.0, 32.3 and 11.2% of screening replies from pass to fail, against at most 4.0% for any other head of their layer, and lowers the pass rate on the crosslingual prompts outside the screen by 0.711, 0.336 and 0.073 (Appendix C), with little monolingual change.

Under mean ablation, each of the ten heads still lowers the pass rate of the task it affects most, keeping 39–89% of the zero-ablation drop with the 95% CI below zero (Appendix C). Across 14 target languages, the selected head reduces crosslingual pass rates in every language in five of six tested models, and in all but Japanese and Russian in Qwen2.5-1.5B (Appendix C.1).

4.3 Specificity Checks

General degradation. On plain text, the seven FLORES heads raise the loss by 0.026 to 0.720 nats, against a layer mean of at most 0.022 for the other heads, so their removal is not free. All seven keep every English continuation in English, beyond the 90% the selection requires. The ten heads also leave English LCB prompts, which play no part in selection, at pass rates of 0.990–1.000. In five models, continuations switching to English stay closer to the prompt in embedding space (0.558–0.683) than the next sentence of the same article (0.384; Appendix D). These checks argue against broad generation failure but do not establish that output quality is preserved.

Head specificity and detectors. Random same-layer heads change LCB pass rates by at most 0.038. Replacing the head's output with a random vector of the same norm moves at least as many continuations as zeroing it (Appendix G), so $C \to W$ alone does not show that the head's output carries language-specific information; in the four models where another head of the layer has a larger output norm, zeroing that head moves at most 1.4%. Relabeling with langid, fastText or a two-of-three vote changes FLORES $C \to W$ by at most 0.02 for all seven heads.

Found again on LCB alone. The same LCB screen, run on the eight models with a FLORES follow-up (with monolingual prompts for the two Gemma models, whose crosslingual baselines of 0.118 and 0.133 leave little room for a drop), ranks the FLORES head first among all eligible heads in five of the seven counted ones (Appendix C). In Qwen2.5-1.5B it ranks second, behind L16H9, which reduces crosslingual pass rates by 0.459 but leaves monolingual replies unchanged; in Qwen2.5-7B it ties for fourth, behind L0H1, whose effect vanishes under mean ablation.

5 Base Models and Post-Training Checkpoints

Do these heads already matter before post-training? We zero the seven FLORES heads in the corresponding base models on the same 2,500 prompts, given as plain text (Base in Table 1). Every head has a smaller effect there: 0.001 to 0.202, or 0.2–69% of the instruction-tuned value. The base models keep the prompt language in 93–99% of unablated continuations, and counting only those continuations changes their values by at most 0.003 (Appendix E), but the comparison is confounded by input format, since only the instruction-tuned models receive a chat template.

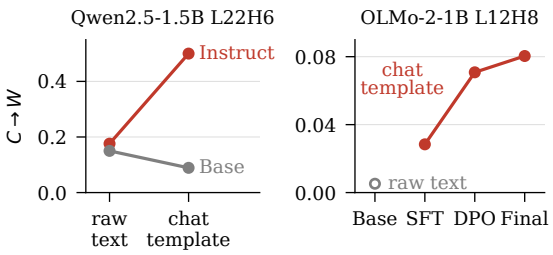


Figure 2: FLORES $C \to W$ on 2,500 prompts when the selected head is zeroed. Left: Qwen2.5-1.5B L22H6 in the Base and Instruct checkpoints, on raw text and with the Instruct chat template. Right: OLMo-2-1B L12H8 in the released checkpoints (Base on raw text, the others with the chat template). The y-axis ranges differ.

Input format in Qwen2.5-1.5B. Crossing checkpoint and input format (Figure 2, left), zeroing L22H6 moves 17.6% (Instruct) and 15.0% (Base) of continuations out of the prompt's language on raw text, but 50.0% (Instruct) and 8.9% (Base) with the Instruct chat template. The head already matters before post-training, and its effect is largest when the Instruct checkpoint receives its own template, suggesting an interaction between post-training and input format.

OLMo-2 checkpoints. Across the released OLMo-2-1B checkpoints (Figure 2, right), the FLORES effect of L12H8 grows from 0.005 (base) to 0.028 (SFT), 0.071 (DPO) and 0.080 (final), while its crosslingual LCB effect is already −0.247 after SFT (Appendix E). The SFT, DPO and final checkpoints share a chat template, so the later increases are not due to format alone, but the base-to-SFT step remains confounded by format, and the final checkpoint adds reinforcement learning with verifiable rewards after DPO, so the checkpoints show how the dependence changes along the released training sequence without isolating its cause.

6 What the Head Does

6.1 Timing, Attention and Values

When the head matters. We first test Qwen2.5-1.5B L22H6 on 96 monolingual LCB prompts (24 per language), sampled independently of earlier ablation results; none names a reply language, and

	Qwen2.5-1.5B		Gemma-3-1B	
Attention from the last prompt token				
To the name		0.773		0.132
To as many nearby tokens		0.003		0.009
Control head, to the name		0.039		0.093
	Fail	Other	Fail	Other
Name masked	34	0	32	0
Nearby tokens masked	0	0	0	0
Control head's mask	0	0	2	0
Head zeroed	105	0	36	0
Same-language values	1	0	0	0
Other-language values	90	44	43	24
Random-direction values	4–8	0	1–6	0
Second-language values	217	208	25	16
L11H2's name masked	–	–	2	0

Table 2: Mechanism checks for Qwen2.5-1.5B L22H6 and Gemma-3-1B L11H3 on 480 crosslingual LCB prompts that name the requested language. The control head attends most to the name among the other heads of the layer (L22H7, L11H1). Values: the values the head reads at the name, replaced by a same-, other- or second-language mean, or moved by the other-language distance in a random direction (three seeds; Other counts German or French). All interventions act in the decoding steps after the first token. Fail: baseline-passing replies (323 and 51) that fail; Other: replies entirely in the replacement language, of the 471 to 474 scored in each condition. Per-sample results in Appendix F.

all pass at baseline. Zeroing the head only in the decoding steps after the first generated token makes 31 replies fail, against 2 when it is zeroed only during prompt processing and 35 when it is zeroed in both; zeroing L22H8, another head of the layer, in both makes none fail. The head thus matters mainly while the reply is generated. All 24 Italian replies fail when it is zeroed during decoding, as in the full LCB run (Italian monolingual LPR 1.00 to 0.00).

Attention to the language name. We then test both heads on 480 crosslingual LCB prompts (120 per language) that name the requested language in English, sampled without using any model's replies: 96 first, then 384 more in a replication fixed before it was run (Appendix F). At the last prompt position, both heads attend to the name more than any other head of their layer, and far more than to as many nearby tokens (Table 2); the runner-up (L22H7, L11H1, chosen before any intervention) serves as the control head. Over the first eight reply tokens, L22H6 still leads, whereas in Gemma-3-1B L11H2 attends slightly more than L11H3 (0.079 against 0.064; Appendix F). Masking the head's attention to the name in the decoding steps after the first generated token makes 34 of 323 baseline-passing Qwen2.5-1.5B replies and 32 of 51 Gemma-3-1B replies fail, while masking its attention to the nearby tokens, or the control head's attention to the name, makes at most two fail. Prespecified two-sided exact McNemar tests (McNemar, 1947) against each control give $p = 0.031$ and $p = 0.016$ on the first 96 prompts (6 of 62 and 7 of 12 replies, uncorrected) and $p < 10^{-6}$ on the 384 new ones. Masking L11H2's attention to the name makes only 2 of the 51 Gemma-3-1B replies fail ($p < 10^{-8}$ against L11H3; Appendix F).

Zeroing the head in the same steps makes 105 and 36 replies fail, including 33 of the 34 and 24 of the 32 mask failures: masking reproduces part of the effect of zeroing the head, and the 31 monolingual failures suggest that the head does more than read the name. At most four baseline-failing replies pass under any of these conditions.

What the name carries. The name mask also removes far more attention than either control, so it does not separate the identity of the name from the amount of attention removed. We therefore leave the head's attention weights as computed and replace only the values it reads at the name with their mean over other prompts that name the same language or another language (German, or French for German prompts). The same-language values make 1 and 0 baseline-passing replies fail, while the other-language values make 90 and 43 fail and put 44 and 24 of all scored replies entirely in the other language, against none under the same-language values ($p < 10^{-6}$ for each comparison, exact McNemar tests fixed before the main runs). The other-language values change the head's output more (mean norm of the change 2.0 against 0.4 in Qwen2.5-1.5B, 0.36 against 0.10 in Gemma-3-1B), so we added two controls, with their readings fixed before they were run (Appendix F). Moving each name value by the same distance in a random direction changes the output at least as much (2.03 to 2.04 and 0.37) but makes only 4 to 8 and 1 to 6 replies fail across three seeds ($p < 10^{-10}$ against the other-language values) and moves none into German or French. Values from a second language (Spanish, Italian or French) put 208 and 16 replies entirely in that language, against none under the same-language values ($p < 10^{-4}$) and at most 8 when the head is zeroed. The values the head reads at the name thus carry information about the language they name into the reply.

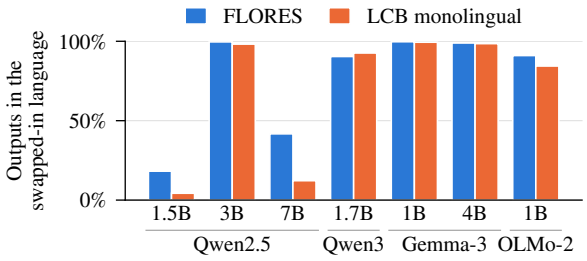


Figure 3: Share of outputs in the swapped-in language when the head's output is replaced by its FLORES mean for another language (German, or French on German prompts). FLORES: the 2,000 non-English continuations, LCB monolingual: replies to the non-English monolingual prompts that are entirely in that language. Exact values in Table 9.

6.2 Replacing the Output with Language Means

We separate redirecting replies into another language from restoring the language a user requested.

Swapping the language. In four models, mean ablation keeps only 6–22% of the zero-ablation effect on FLORES (Table 1). Replacing the output at every position with another language's mean $\mu_{h,\ell}$ (Section 3.2) instead redirects the output language in five of the seven models (Figure 3): 90.4–99.7% of the 2,000 non-English FLORES continuations, on the prompts used to compute the means, are then in the swapped-in language, while the prompt language's own mean moves at most 0.1% out of it. With the same means, computed on FLORES only, 84.4–99.4% of the replies to non-English monolingual LCB prompts are entirely in that language, and swapping any of three same-layer control heads per model puts none there. Qwen2.5-1.5B and Qwen2.5-7B show far lower rates (18.2% and 41.7% on FLORES, 4.3% and 12.2% on LCB).

Steering toward the requested language. We count replacing the output with the mean for the language a crosslingual LCB prompt requests as an improvement when the lower end of the 95% CI of the pass rate change is above zero and above every control head's change. This holds only in the two Gemma models, whose crosslingual baselines are low: from 0.118 to 0.560 in Gemma-3-1B (paired ΔLPR $+0.440$ [$+0.410$, $+0.469$]) and from 0.133 to 0.213 in Gemma-3-4B ($+0.079$ [$+0.064$, $+0.095$]). The other five start at 0.70 to 0.95, and in Qwen2.5-1.5B, at 0.704, the pass rate falls by 0.020. Setting the head's output can thus redirect the reply language, but it does not reliably restore a requested one (content of the corrected replies and further diagnostics in Appendix G).

7 Discussion and Conclusion

We ablated attention heads one at a time in 12 instruction-tuned models. In ten of them, removing one head moves far more outputs out of the expected language than removing any other head of its layer, and its LCB effect persists, though reduced, under mean ablation. Which task the head affects varies: some heads stand out on FLORES continuations, others mainly on explicitly requested languages, and in Qwen2.5-1.5B and Qwen2.5-7B the LCB screen favors a different head from the FLORES screen. In an exploratory comparison, the dependence is weaker in most larger models: the largest LCB drop per model is 0.28 to 0.71 in the seven counted 1–3B models, whereas in three of the four models with 4B or more parameters the strongest head lowers LCB pass rates by at most 0.076, a comparison that does not separate size from model family.

The seven FLORES heads matter less in the base models given plain text, and the Qwen2.5-1.5B format runs and the OLMo-2-1B checkpoints suggest that this dependence changes with post-training and prompt format. A head's importance, its mechanism and its use for steering are related but distinct. In two models, the values the head reads at the name of the requested language carry information about that language into the reply, beyond the size of the change, yet the name is not its only input. Replacing the head's output with another language's mean redirects the output language in five models, but Qwen2.5-1.5B L22H6, with one of the largest ablation effects, redirects only 18.2% of FLORES continuations, and the requested language's mean meets our criterion for improving crosslingual replies only in the two Gemma models. A head can thus be important for keeping the reply language without being a reliable way to set it.

Limitations

Evaluation coverage. FLORES covers five European languages, and the main LCB evaluation four requested non-English languages plus English controls; the 14-language extension covers six of the twelve models. The results therefore do not establish generality across languages, scripts or multilin-

gual settings. LCB's line-level metric leaves replies without a line of at least five words unscored, which can omit short answers and some script switches where words are not separated by spaces. LPR measures language adherence, not whether a reply is correct.

Scope of the single-head claim. We test models of 1B–7B parameters, and the effects are smaller in most of those with 4B or more. Our procedure reports at most one head per model and does not ablate sets of heads, so it cannot show how much of the function other heads share or recover through self-repair (McGrath et al., 2023). We induce departures by intervention and do not test whether naturally occurring language confusion involves the same heads, and the five FLORES languages are all European languages in Latin script.

Screening and measurement. Heads are selected by screening, so the results describe the selected heads, not how common such heads are across models. The 125-prompt FLORES screen can miss rare or prompt-specific effects, and selecting heads by their observed effect favors upward noise; the seven FLORES heads still reach 0.077–0.517 on the 2,375 prompts outside the screen, although these prompts also enter the final selection. The decision to leave OLMo-3-7B's L14H25 out of the count was made after its LCB results were in. The ΔNLL and English-retention checks limit, but cannot exclude, general damage to generation; language detection is imperfect despite the detector checks; and effects are compared within a layer, not across architectures.

Intervention semantics. Zeroing a head before the output projection can change downstream computation, including other heads, and where the attention output is normalized after the projection (Gemma-3, OLMo-2, OLMo-3) the norm rescales the change; holding it fixed accounts for part of the effect in Gemma-3-1B. Mean ablation and language-specific replacement insert fixed activations that may be atypical for a given prompt or position, so neither isolates a semantic feature on its own.

Precision and checkpoints. Qwen2.5-1.5B runs in fp16 (fp32 in Section 6.1), Qwen2.5-7B and OLMo-3-7B in bf16, and the other models in fp32. In bf16, batched and single-prompt runs produce identical continuations on only 15 of 20 checked prompts in Qwen2.5-7B and 10 of 20 in OLMo-3-7B, so small effects in these two models call for caution. OLMo-2-1B Base was screened in bf16, with its selected layer rerun in fp32, and its FLORES continuations block the end-of-text token. Evaluations use greedy decoding, with the repetition penalty set in Qwen2.5-Instruct's generation configuration, and do not cover sampling. The base and instruction-tuned comparisons use released checkpoints rather than controlled training, and base models get raw text while instruction-tuned ones get their chat template; the Qwen2.5-1.5B format comparison separates the two only in part, so the comparisons cannot attribute a change to any post-training stage.

Mechanism and steering. The mechanism checks cover two models and 480 crosslingual prompts, with 51 baseline-passing replies for Gemma-3-1B. The random-direction and second-language controls separate what the name carries from the size of the change and from a bias toward one replacement language, but they were added after the first value-patch results and test two other languages per prompt language. The mask reproduces only some of the failures from zeroing the head, so the head has other inputs. FLORES swap rates are measured on the prompts used to estimate the language means, while LCB tests transfer to separate prompts. Steering meets our improvement criterion only in the two Gemma models and does not establish robustness in deployment, that response content is preserved, or a general way to reduce language confusion.

Responsible use. The study uses publicly available models and benchmarks. Head-level interventions can deliberately change a model's output language and may alter response content, so practical use would require model-specific validation and safeguards against unintended changes.

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A Supplementary Protocol Details

Each instruction-tuned input contains a single user turn, rendered with the model's own chat template. Thinking is disabled where supported; templates that insert a date use 2026-10-08. Qwen2.5 uses its default English system prompt, and its generation configuration adds a repetition penalty (1.1 for 1.5B, 1.05 for 3B and 7B) to every greedy run. FLORES continuations use greedy decoding with up to 40 new tokens, and LCB replies with up to 100. The FLORES screen uses 25 prompts per language: the five hand-written starts and the first 20 filtered FLORES sentences. A model proceeds to the 2,500-prompt follow-up if at least one head meets both the loss and English-retention conditions and reaches $C \to W \geq 0.10$ on the screen (Section 3.4). The FLORES sentences are the first 495 aligned devtest sentences, in dataset order, for which the versions in seven languages (the five prompt languages, Chinese and Russian) all contain 20 to 300 characters and at least half of the characters are letters. `langdetect` runs with its random seed fixed to zero, and a continuation it cannot label counts as $\neq \ell_j$ (one continuation across the seven heads found on FLORES).

For LCB, paired bootstrap confidence intervals compare an intervention and the baseline on the same prompts; the percentile interval of 2,000 resamples is not stratified by source. The benchmark's line-level scoring ignores lines shorter than five words, counts a fastText label with confidence of 0.3 or less as unknown, and averages pass rates over prompt sources, with the four languages pooled within each source. ΔLPR is computed over 778 to 800 of the 800 non-English monolingual and 1,159 to 1,190 of the 1,196 crosslingual prompts for the heads in Table 1. Chinese and Japanese are segmented before the five-word filter in the 14-language extension, as in the benchmark's scoring script (Marchisio et al., 2024). Each run uses one NVIDIA H100 80GB GPU. We use fp16 for Qwen2.5-1.5B (fp32 in Section 6.1), bf16 for OLMo-3-7B and Qwen2.5-7B, and fp32 for the other models; each base checkpoint uses the precision of its instruction-tuned model, except that the OLMo-2-1B base screen ran in bf16 with its selected layer rerun in fp32. The OLMo-2-1B base run also blocks the end-of-text token, which it otherwise emits right after most complete FLORES sentences.

B FLORES Screens and Full-Set Follow-ups

Table 3 summarizes the FLORES screens of the 12 instruction-tuned models. The reported $C \to W$ and mean-ablation rates are from the 2,500-prompt follow-up where one was run. The 0.10 threshold applies to the 125-prompt screen, so a full-set value can fall below it: OLMo-2-1B's L12H8 reaches 0.136 on the screen and 0.080 on the full set. Parentheses in the ΔNLL column give the mean change for the other heads in the selected layer. The follow-up reruns every head of the layers that hold the two heads with the highest screened $C \to W$, whether or not they meet the conditions, and the head with the highest screened $C \to W$ among those with $\Delta\text{NLL} \leq 0.1$ (all layers for Qwen2.5-1.5B). Outside the rerun layers, no head that meets both conditions exceeds a screened $C \to W$ of 0.064. Any English threshold from 0.6 to 0.95 selects the same head in each of the eight models with a follow-up. On the 2,375 prompts not used in the screen, the seven heads' $C \to W$ is 0.077 to 0.517, close to 0.080 to 0.515 on all 2,500 and higher than on the 125 screen prompts in four of the seven models. A dash marks a value not obtained because no FLORES head met the screening rule; the heads found on LCB are in Table 4.

Seven heads found on FLORES are counted in the main analysis. OLMo-3-7B L14H25 is not: its LCB pass-rate changes are small (-0.041 on monolingual and -0.018 on crosslingual prompts) and vanish under mean ablation ($+0.001$ and $+0.015$; Table 5), whereas every counted head keeps a drop with its 95% CI below zero on at least one LCB task. The separate LCB screen also finds no comparably distinct head in OLMo-3-7B (largest drop 0.060). This criterion was set after the LCB evaluation. Its small FLORES mean-ablation effect (0.007) does not decide the question on its own, since mean ablation also keeps only 6–22% of the FLORES effect in Qwen2.5-7B, Qwen3-1.7B and the two Gemma models (Appendix G). Screening rates are not used to rank heads across model families, whose head counts and normalization differ.

C LCB Results Across Models

Table 4 gives the change in line-level pass rate under zero ablation with its 95% paired-bootstrap interval, next to the changes for the same-layer control heads. Across models, the control heads change monolingual LPR by -0.011 to $+0.010$

Model	FLORES head ($C \to W$)	Mean	ΔNLL (layer mean)	Status	Note
Qwen2.5-1.5B	L22H6 (0.500)	0.450	+0.228 (+0.004)	Follow-up	Base counterpart: 0.150
Qwen2.5-3B	L27H13 (0.515)	0.338	+0.234 (+0.006)	Follow-up	Base counterpart: 0.202
Qwen2.5-7B	L19H1 (0.200)	0.018	+0.026 (+0.001)	Follow-up	L0H25 (0.629) and L0H22 (0.466) fail English retention (0.512 and 0.518)
Qwen3-1.7B	L18H12 (0.324)	0.028	+0.128 (+0.003)	Follow-up	L0H3 (0.466) fails the loss condition ($\Delta\text{NLL} > 1$)
Qwen3-4B	None (screen 0.024)	–	–	Screen only	LCB screened directly
Gemma-3-1B	L11H3 (0.413)	0.089	+0.720 (−0.012)	Follow-up	Same head has little effect in Base
Gemma-3-4B	L24H0 (0.214)	0.012	+0.202 (−0.005)	Follow-up	Base counterpart: 0.148
OLMo-2-1B	L12H8 (0.080)	0.056	+0.136 (+0.022)	Follow-up	Small effect; also observed on LCB
OLMo-3-7B	L14H25 (0.134)	0.007	+0.202 (+0.000)	Follow-up	bf16; small LCB effect, no distinct head on LCB screen
Llama-3.2-1B	None (screen 0.048)	–	–	Screen only	LCB-only head: L8H25
Llama-3.2-3B	None (screen 0.008)	–	–	Screen only	LCB-only head: L13H19
SmolLM3-3B	None	–	–	Screen only	L1H12 reaches 0.880 but has ΔNLL +2.58 and repeated `</think>` output

Table 3: FLORES head screens and full-set follow-ups. Status: Follow-up, the model went on to the 2,500 prompts; Screen only, no head met the screening rule, so only the 125-prompt screen was run (screen values are the strongest result there, not full-set estimates). Mean: $C \to W$ under pooled mean ablation. OLMo-3-7B ran the follow-up but is not counted (see text).

and crosslingual LPR by -0.025 to $+0.038$. Gemma's crosslingual baseline LPR is already low (0.118 for 1B and 0.133 for 4B), which limits the additional drop these models can show on that task.

The direct LCB screen ablated every head on 100 crosslingual prompts. For the three heads selected through this route, the unweighted share of replies scored in both conditions that changed from pass to fail was 0.710 for Llama-3.2-1B L8H25 (71/100), 0.323 for Llama-3.2-3B L13H19 (32/99), and 0.112 for Qwen3-4B L24H27 (11/98). The largest such share for another head of the same layer was 0.040. These values count pass-to-fail transitions rather than net changes in pass rate; in Qwen3-4B, 0.010 changed from fail to pass, giving a net change of -0.102 on the screen. The corresponding crosslingual changes on prompts outside the screen were -0.711, -0.336, and -0.073 (95% CI $[-0.090, -0.057]$ for Qwen3-4B). SmolLM3-3B yielded no eligible head. These checks distinguish the LCB-selected heads from the other heads in their layers and show effects on prompts that were not used to select them.

Table 5 repeats the LCB evaluation with mean ablation. Each of the ten counted heads keeps a drop with its 95% CI below zero on the task it affects most, 39–89% of the zero-ablation drop, while OLMo-3-7B L14H25 does not. The same table gives the two heads that the LCB screen selects instead of the FLORES head (Section 4.3): Qwen2.5-1.5B L16H9 keeps its crosslingual drop under mean ablation, and Qwen2.5-7B L0H1 does not.

On that screen, run on the eight models with a FLORES follow-up, the FLORES head lowers the pass rate on the 100 screening prompts by 0.389 in Qwen2.5-3B, 0.485 in Qwen3-1.7B, 0.343 in OLMo-2-1B, 0.660 in Gemma-3-1B and 0.340 in Gemma-3-4B (the two Gemma models on monolingual prompts), the largest drop among the heads that meet both conditions in each, against 0.052, 0.051, 0.081, 0.030 and 0.030 for the next head. In Qwen2.5-1.5B, L22H6 lowers it by 0.220, second of 336 heads, behind L16H9 (0.400); in Qwen2.5-7B, L19H1 lowers it by 0.040, tied for fourth of 780 heads that meet both conditions, behind L0H1 (0.232). OLMo-3-7B is described in Appendix B.

C.1 Fourteen-language LCB extension

We extended LCB to its 14 non-English languages (Arabic, Chinese, French, German, Hindi, Indonesian, Italian, Japanese, Korean, Portuguese, Russian, Spanish, Turkish and Vietnamese) for six models: Qwen2.5-1.5B, Qwen2.5-3B, Qwen3-1.7B, Gemma-3-1B, OLMo-2-1B, and Llama-3.2-1B. Under the predefined criterion that a language's paired interval is below zero and its change is below every same-layer control, the selected head lowers crosslingual LPR in all 14 languages for five models and in 12 of 14 for Qwen2.5-1.5B. Monolingual effects vary more: 10/14 for Qwen2.5-1.5B, 13/14 for Qwen2.5-3B, 12/14 for Qwen3-1.7B, 14/14 for Gemma-3-1B, 13/14 for OLMo-2-1B, and 6/14 for Llama-3.2-1B. The Qwen2.5-1.5B exceptions

		Selected head		Same-layer controls	
Model	Head	Δ mono LPR	Δ cross LPR	Δ mono	Δ cross
Qwen2.5-1.5B	L22H6	−0.273 [−0.305, −0.240]	−0.276 [−0.303, −0.250]	−0.001 to +0.004	−0.007 to +0.000
Qwen2.5-3B	L27H13	−0.537 [−0.575, −0.503]	−0.449 [−0.478, −0.421]	+0.000 to +0.003	−0.025 to +0.004
Qwen2.5-7B	L19H1	−0.009 [−0.020, +0.003]	−0.074 [−0.089, −0.059]	+0.001 to +0.003	−0.008 to +0.001
Qwen3-1.7B	L18H12	−0.149 [−0.174, −0.123]	−0.508 [−0.535, −0.478]	−0.004 to −0.003	+0.000 to +0.021
Qwen3-4B	L24H27	+0.000 [−0.010, +0.010]	−0.076 [−0.093, −0.061]	−0.005 to +0.010	−0.001 to +0.013
Gemma-3-1B	L11H3	−0.628 [−0.662, −0.593]	−0.091 [−0.110, −0.073]	−0.005 to +0.003	−0.023 to +0.038
Gemma-3-4B	L24H0	−0.259 [−0.292, −0.227]	−0.107 [−0.126, −0.090]	−0.001 to +0.000	+0.000 to +0.012
OLMo-2-1B	L12H8	−0.272 [−0.302, −0.239]	−0.335 [−0.362, −0.308]	+0.000 to +0.004	−0.003 to +0.004
OLMo-3-7B	L14H25	−0.041 [−0.060, −0.024]	−0.018 [−0.033, −0.003]	−0.011 to +0.004	+0.001 to +0.007
Llama-3.2-1B	L8H25	−0.016 [−0.026, −0.008]	−0.711 [−0.737, −0.684]	−0.009	+0.001
Llama-3.2-3B	L13H19	+0.000 [−0.006, +0.006]	−0.335 [−0.363, −0.308]	+0.000 to +0.006	−0.003 to +0.000

Table 4: LCB pass-rate change under zero ablation, with 95% paired-bootstrap intervals. Same-layer controls: range over three other heads of the layer (one, L8H12, for Llama-3.2-1B). The Llama-3.2-1B, Llama-3.2-3B and Qwen3-4B heads were found by screening LCB directly, the others on FLORES.

Model	Head	Δ mono LPR	Δ cross LPR
Qwen2.5-1.5B	L22H6	−0.246 [−0.276, −0.215]	−0.247 [−0.273, −0.220]
Qwen2.5-3B	L27H13	−0.409 [−0.444, −0.373]	−0.331 [−0.358, −0.303]
Qwen2.5-7B	L19H1	+0.003 [−0.005, +0.010]	−0.035 [−0.047, −0.024]
Qwen3-1.7B	L18H12	−0.067 [−0.086, −0.048]	−0.230 [−0.257, −0.205]
Qwen3-4B	L24H27	−0.004 [−0.014, +0.006]	−0.040 [−0.052, −0.027]
Gemma-3-1B	L11H3	−0.244 [−0.275, −0.213]	−0.092 [−0.110, −0.075]
Gemma-3-4B	L24H0	−0.120 [−0.145, −0.095]	−0.053 [−0.066, −0.040]
OLMo-2-1B	L12H8	−0.128 [−0.152, −0.104]	−0.143 [−0.165, −0.123]
Llama-3.2-1B	L8H25	−0.015 [−0.025, −0.005]	−0.357 [−0.384, −0.329]
Llama-3.2-3B	L13H19	−0.005 [−0.011, +0.001]	−0.250 [−0.277, −0.225]
OLMo-3-7B	L14H25	+0.001 [−0.011, +0.015]	+0.015 [+0.003, +0.028]
Qwen2.5-1.5B	L16H9	+0.008 [−0.003, +0.018]	−0.277 [−0.303, −0.251]
Qwen2.5-7B	L0H1	+0.004 [−0.005, +0.013]	−0.005 [−0.013, +0.003]

Table 5: LCB pass-rate change under mean ablation, with 95% paired-bootstrap intervals. Below the line: the uncounted OLMo-3-7B head and the two heads the LCB screen selects instead of the FLORES head; under zero ablation, L16H9 changes the pass rates by -0.009 and -0.459, and L0H1 by -0.116 and -0.271.

on crosslingual prompts are Japanese and Russian; its language-specific pattern does not transfer uniformly to the other models. These are per-language comparisons, and the intervals are not adjusted for the number of languages.

D Detector and Content Checks

Alternative language detectors. We relabeled the full-set FLORES continuations with langid (Lui and Baldwin, 2012), with fastText (Joulin et al., 2016), and by a vote that takes a label when at least two of langdetect (Nakatani, 2010), langid and fastText agree. Across all seven heads found on FLORES, $C \to W$ differs by at most 0.02 from the langdetect result. Under every detector, the selected head ranks first among the heads rerun on 2,500 prompts, except in Qwen3-1.7B, where L18H12 ranks second to L0H3, and Qwen2.5-7B, where L19H1 ranks third after L0H25 and L0H22. These three higher-ranked heads fail the head-selection conditions. For Qwen2.5-7B L19H1, $C \to W$ is 0.200 with langdetect and 0.200–0.208 with the alternative detectors.

Content similarity after a switch. For prompts whose non-English continuation switches to English after ablation, we measured cosine similarity between the prompt and the continuation using Qwen3-Embedding-0.6B (Zhang et al., 2025). For scale, over all non-English FLORES prompts, not only the switched ones, the next sentence of the same article has similarity 0.384 to the prompt (0.370 when that sentence is taken in English), and a random sentence 0.179 (0.172 in English).

The departures do not always go to English. With the head zeroed, 918 of Qwen2.5-1.5B's 1,250 departures go to English, but Qwen2.5-3B's 1,288 split into English (514), Chinese (299) and Spanish (221), and 441 of Qwen2.5-7B's 500 go to Chinese. In Qwen3-1.7B and Gemma-3-1B, the

Model	Switched prompts	Before	After	Before/after reply
Qwen2.5-1.5B	918	0.475	0.558	–
Qwen2.5-3B	514	0.590	0.657	0.582
Qwen3-1.7B	739	0.635	0.683	0.600
Gemma-3-1B	873	0.639	0.559	0.586
OLMo-2-1B	101	0.702	0.668	0.607

Table 6: Content of the continuations that switch to English when the head is zeroed. Switched prompts: their number, Before and After: cosine similarity between the prompt and the continuation without and with the ablation, Before/after reply: similarity between the two continuations, –: not measured. Same-layer control heads switch at most two prompts each.

739 and 873 continuations that switch to English (Table 6) are most of the departures ($C \rightarrow W$ of 0.324 and 0.413 on 2,500 prompts).

In the five models, switched English continuations have a mean similarity of 0.558–0.683 to their prompts, above the reference values for the next sentence of the same article (0.384 in the original language and 0.370 in English). Because the references are computed on all non-English prompts rather than on the switched ones, the comparison is only a rough indication of content relatedness; it does not establish equivalent meaning or task correctness.

E Base and Post-training Checkpoints

For the seven heads found on FLORES, the base-model values are in the Base column of Table 1; each head has a smaller full-set $C \rightarrow W$ in the base model. Counting only the base continuations that stay in the prompt's language without ablation changes the base-model values by at most 0.003. The OLMo-2-1B comparison follows L12H8 across the released base, SFT, DPO, and instruction-tuned checkpoints (Team OLMo et al., 2024), with FLORES $C \rightarrow W$ of 0.005, 0.028, 0.071, and 0.080. Base layer 12 is run in fp32, as the later checkpoints are. Crosslingual LCB changes under zero ablation are -0.247, -0.244, and -0.335 for SFT, DPO, and the final checkpoint. The base checkpoint is tested without a chat template and with the end-of-text token blocked. The final checkpoint adds reinforcement learning with verifiable rewards after DPO, released through an intermediate RLVR checkpoint, so the results do not isolate the effects of individual training stages.

F Mechanism Check Details

Setup. The crosslingual prompts are drawn with a fixed seed from the LCB crosslingual prompts that name the requested language in English, round robin over sources, without using any model's replies. The 96 monolingual prompts for the phase experiment are drawn from Qwen2.5-1.5B's baseline LCB rows with a fixed seed, without filtering on earlier ablation results; the Italian and German prompts all come from Okapi (Lai et al., 2023), and the French and Spanish prompts eight each from Dolly (Conover et al., 2023), the native set and Okapi. Both runs use fp32, eager attention, greedy decoding and 100 new tokens. A mask adds $-\infty$ to the attention logits of the selected head only, at the masked key positions, in every decoding step after the first generated token; the first generated token matches the clean run in every condition. The nearby tokens are as many user-text tokens as the language name has, the ones closest to it. The control head is the other head of the layer with the largest mean attention from the last prompt token to the name in a clean pass, chosen before any intervention. Attention is measured by teacher forcing the clean reply. Over the first eight reply tokens of the 480 prompts, L22H6 puts 0.375 of its attention on the name, against 0.018 for L22H7 and 0.001 for the nearby tokens; L11H3 puts 0.064, against 0.020 for L11H1, 0.079 for L11H2 and 0.005 for the nearby tokens. The exact McNemar tests (McNemar, 1947) compare the name mask with each control on the baseline-passing replies. The reading fixed in advance requires the name mask to fail more replies than both controls in a model, so the two tests per model form an intersection-union test (Berger, 1982) that needs no correction; Holm's correction (Holm, 1979) across the two models gives $p = 0.031$ for both.

Earlier selected-prompt check. An earlier Qwen2.5-1.5B L22H6 run used 96 crosslingual LCB prompts, including Italian prompts selected from replies that had switched language under full-head ablation. On this set, 95 replies passed at baseline, mean attention to the requested-language name was 0.807, and masking that attention caused seven baseline-passing replies to fail, against none for nearby tokens or for the name at L22H8. All seven also failed when the head was zeroed in the decoding steps. Because prompt selection depended in part on earlier intervention outcomes, we report this run only as a supporting check.

			Masked				Values replaced		
Model	Prompts	Pass	Name	Near	Ctrl	Zeroed	Same	Other	In other
Qwen2.5-1.5B	96	62	6	0	0	20	0	17	8/95
	384 new	261	28	0	0	85	1	73	36/378
	all 480	323	34	0	0	105	1	90	44/473
Gemma-3-1B	96	12	7	0	0	8	0	10	7/95
	384 new	39	25	0	2	28	0	33	17/376
	all 480	51	32	0	2	36	0	43	24/471

Table 7: Mechanism checks on the 96 earlier and 384 new crosslingual LCB prompts. Pass: replies that pass at baseline, of which the columns from Name to Other count the ones that fail, Name, Near and Ctrl: the head's attention to the language name or to as many nearby tokens masked, or the control head's attention to the name, Zeroed: the head zeroed, Same and Other: the values the head reads at the name replaced by their mean over prompts that name the same or another language, In other: scored replies entirely in that other language, out of all scored replies, all in the decoding steps after the first generated token. The masks on the 96 prompts come from the earlier run, whose clean replies these runs reproduce exactly.

Value patching and a larger sample. Two checks were written down, together with the code that runs them, before they were run: the same mechanism runs on 384 new crosslingual prompts, drawn by the same rule as the 96 of Section 6.1 and leaving those out, and a value patch on all 480 prompts. Because the draw does not use model outputs, 31 of the 384 happen to be among the prompts of the selected-prompt check. The value patch leaves the selected head's attention weights as computed and replaces the values it reads at the name positions: it adds $\sum_j \alpha_{tj}(\bar{v} - v_j)$ over those positions to the head's attention output before the output projection, without changing the shared key-value cache or directly intervening on other heads (the layer has 2 key-value heads for 12 query heads in Qwen2.5-1.5B and 1 for 4 in Gemma-3-1B). $\bar{v}$ is the mean value at the name positions over the other prompts of the run that name the same language, or over the prompts that name the other language (German for French, Spanish and Italian prompts, French for German ones). Patching each name position with its own value left every reply unchanged in a smoke run. The prespecified readings are that the other-language patch fails more baseline-passing replies than the same-language patch and puts more replies entirely in the other language, both by exact McNemar tests on the 480 prompts, unless the same-language patch already fails more replies than the nearby mask. A note added after the smoke run and before the main runs counts the first reading only together with the second, since the other-language patch changes the head's output more. Both criteria are met in both models (Table 7): the other-language patch alone fails 89 and 43 replies that the same-language patch keeps, with no reply the other way ($p \leq 2.3 \times 10^{-13}$), and alone puts 44 and 24 replies in the other language, again with none the other way ($p \leq 1.2 \times 10^{-7}$), and the same-language patch does not fail significantly more replies than the nearby mask (1 against 0 and 0 against 0). No reply is entirely in the other language under the name mask, the nearby mask, the control head's mask or zeroing the head (0 of 471 to 474 scored replies in each model). Few replies change in the other direction: under the other-language patch, 3 baseline-failing Qwen2.5-1.5B replies pass and none in Gemma-3-1B. In Qwen2.5-1.5B, Italian replies again dominate: of 79 baseline-passing ones, the name mask fails 20, zeroing the head 79 and the other-language patch 51.

Controls for the value patch. Because the other-language patch changes the head's output more, we added three controls after the results above, with their readings written down before they were run, together with the code that runs them. The random-direction patch moves each name value v_j by its distance to the other-language donor in a random direction, $v_j + \|\bar{v}_{\text{other}} - v_j\|\, u$, with one unit vector u per prompt shared by its name positions, so that the change to the head's output is at least as large as under the other-language patch at each step; three seeds. The second-language patch uses the donor of a second language (Spanish for German prompts, Italian for Spanish and French prompts, French for Italian prompts). In Gemma-3-1B, we also mask L11H2's attention to the name, the head that attends to it most over the first eight reply tokens. All runs reproduce the earlier clean replies on the

480 prompts. Every planned reading holds in both models (Table 8): the other-language patch fails more replies than each random direction (87, 83 and 88 against 2, 1 and 2 discordant in Qwen2.5-1.5B; 42, 40 and 37 against 0 in Gemma-3-1B; $p \leq 1.5 \times 10^{-11}$) and puts more replies in the first donor language (44 and 24 against 0 for every seed); the second-language patch puts more replies in its language than the same-language patch (205 and 15 against 0; $p \leq 6.1 \times 10^{-5}$); and the L11H3 name mask fails more replies than the L11H2 name mask (30 against 0; $p = 1.9 \times 10^{-9}$). Zeroing the head puts 8 Qwen2.5-1.5B replies entirely in the second donor language and the name mask 4, all Italian prompts answered in French, against none in Gemma-3-1B; no mask and no zeroing puts a reply entirely in the first donor language.

G Zero and Mean Ablation Diagnostics

Mean replacement sometimes affects continuation language much less than zero ablation (Table 1). To examine this difference, we evaluated the seven heads found on FLORES with additional interventions on 2,500 prompts. These include pooled-mean and language-conditioned replacement, mean subtraction, random-vector replacement, and output scaling. For models with post-attention normalization, we also tested a control that holds the clean normalization factor fixed. Table 9 reports $C \rightarrow W$ on the 2,500 FLORES prompts for each intervention. Panel (a) also gives three statistics of the head's projected output c_t over the user text and continuation tokens: the rank of its mean norm in the layer (Norm rank), the share $\|m\|^2/\mathbb{E}\|c_t\|^2$ of its energy in the mean vector m (Mean energy), and the share of the remaining variance explained by the prompt's language, $\sum_\ell n_\ell \|m_\ell - m\|^2 / \sum_t \|c_t - m\|^2$ with m_ℓ the mean over the n_ℓ tokens of language ℓ (Lang. residual). Prompt mean and Cont. mean pool the head's output over prompt tokens and over continuation tokens, and Minus cont. mean subtracts the latter. In panel (b), the own-language, English and other-language means are the language means of Section 3.2, with German for French, Spanish and Italian prompts and French for German ones. Swapped-in language gives the share of the 2,000 non-English FLORES continuations in that language, and of the scored replies to non-English LCB prompts entirely in it. Random same norm replaces the output with a random vector of the same norm, Half output scales it by 0.5, and Zero, norm fixed holds the clean post-attention normalization factor fixed.

Replacing a head with the mean for the prompt's own language produces few correct-to-wrong changes ($C \rightarrow W \leq 0.052$ in all seven models). Replacing it with another language's mean yields higher $C \rightarrow W$ than zero ablation in every model. The choice of replacement mean therefore matters: for Qwen3-1.7B, Gemma-3-1B, and Gemma-3-4B, prompt-token pooled means yield $C \rightarrow W$ rates of 0.028, 0.089, and 0.012, respectively, whereas other-language means yield 0.780, 0.977, and 0.796. Continuation-token pooled means yield higher rates for Qwen3-1.7B (0.155) and Gemma-3-1B (0.196), showing that the pooled-mean result also depends on which tokens are averaged.

$C \rightarrow W$ alone is not specific to a language: a random vector of the same norm moves about as many continuations as another language's mean in Qwen2.5-1.5B (0.544 against 0.530), so the share in the swapped-in language is the measure of redirection. The language means are estimated on the same FLORES prompts used for the FLORES swap, whereas the LCB results test transfer to separate prompts. In five models, at least 0.904 of continuations to non-English prompts under other-language replacement are in the replacement language, and the FLORES means put 0.844–0.994 of the monolingual LCB replies in that language. The shares are lower for Qwen2.5-1.5B and Qwen2.5-7B. Holding the post-attention normalization factor fixed reduces the zero-ablation rate for Gemma-3-1B from 0.413 to 0.160, but changes the rate little for Gemma-3-4B (0.214 to 0.234) and OLMo-2-1B (0.080 to 0.074). A small effect under pooled-mean replacement therefore does not, by itself, show that the head has little influence on continuation language.

Content of replies under replacement. For the crosslingual LCB replies that replacement with the requested language's mean turns from fail to pass, we compute the cosine similarity (Qwen3-Embedding-0.6B) to the baseline reply to the same prompt and to the baseline reply to another prompt of the same task and language, with one fixed pairing per prompt. The corrected replies (530 in Gemma-3-1B, 98 in Gemma-3-4B, 43 in Qwen3-1.7B, 31 in Qwen2.5-3B, 20 in OLMo-2-1B, 16 in Qwen2.5-1.5B and 11 in Qwen2.5-7B) have similarity 0.69–0.86 with the baseline reply to the same

Model	Condition	Fail	First	Second	Change
Qwen2.5-1.5B	Same language	1	0	0	0.391
(323 pass)	Other language	90	44	2	2.007
	Random, seed 1	5	0	0	2.034
	Random, seed 2	8	0	0	2.028
	Random, seed 3	4	0	0	2.043
	Second language	217	0	208	1.976
Gemma-3-1B	Same language	0	0	0	0.102
(51 pass)	Other language	43	24	0	0.360
	Random, seed 1	1	0	0	0.367
	Random, seed 2	3	0	0	0.367
	Random, seed 3	6	0	0	0.368
	Second language	25	0	16	0.335
	L11H2 name mask	2	0	0	–

Table 8: Value patch controls on the 480 crosslingual prompts. Fail: baseline-passing replies that fail; First and Second: scored replies (471 to 476) entirely in the first donor language (German, or French for German prompts) or the second (Spanish, Italian or French); Change: mean norm of the change to the head's output per decoding step.

prompt, against 0.15–0.21 with another prompt's for the replies whose pass status changes. Under this replacement, no model's share of unscored replies or of repetitive replies rises by more than 0.002. Similarity indicates overlapping content, not the same meaning.

H Comparison with Translation Heads

Lasnier et al. (2026b) identify *language heads*, which select the output language in few-shot translation, by activation patching on base models. We compare our instruction-tuned heads with the per-head scores released by the authors for Qwen3-1.7B-Base, Llama-3.2-1B, and Gemma-3-1B-pt (Lasnier et al., 2026a, commit `daa721b`). The scores cover 20 English-to and English-from directions involving French, Spanish, Portuguese, Japanese, Chinese, Swahili, Wolof, Hindi, Arabic, and Russian. We average each direction over its examples, count how often our head is the direction's top head, and rank its mean score across directions (Table 10).

The match is strongest for Qwen3-1.7B and Llama-3.2-1B, but it does not show that the heads serve the same function in base-model translation and instruction-tuned language retention. Gemma-3-1B L11H3 ranks third by mean score; another head, L15H2, tops all 20 directions.

(a) Head-output statistics and pooled-mean interventions

Model	Head	Norm rank	Mean energy	Lang. residual	Zero	Prompt mean	Cont. mean	Minus cont. mean
Qwen2.5-1.5B	L22H6	1/12	0.18	0.62	0.501	0.451	0.431	0.027
Qwen2.5-3B	L27H13	2/16	0.38	0.71	0.515	0.338	0.372	0.059
Qwen2.5-7B	L19H1	2/28	0.38	0.77	0.202	0.021	0.008	0.070
Qwen3-1.7B	L18H12	1/16	0.34	0.56	0.324	0.028	0.155	0.138
Gemma-3-1B	L11H3	1/4	0.76	0.03	0.413	0.089	0.196	0.070
Gemma-3-4B	L24H0	2/8	0.56	0.73	0.214	0.012	0.011	0.003
OLMo-2-1B	L12H8	2/16	0.31	0.67	0.080	0.056	0.054	0.005

(b) Language-conditioned and control interventions

				Swapped-in language					
Model	Own-lang. mean	English mean	Other-lang. mean	FLORES	LCB mono	LCB cross	Random same norm	Half output	Zero, norm fixed
Qwen2.5-1.5B	0.052	0.486	0.530	0.182	0.043	0.019	0.544	0.071	–
Qwen2.5-3B	0.001	0.595	0.794	0.996	0.982	0.829	0.636	0.047	–
Qwen2.5-7B	0.000	0.188	0.352	0.417	0.122	0.128	0.228	0.017	–
Qwen3-1.7B	0.000	0.667	0.780	0.904	0.926	0.701	0.377	0.009	–
Gemma-3-1B	0.000	0.782	0.977	0.997	0.994	0.594	0.666	0.008	0.160
Gemma-3-4B	0.000	0.222	0.796	0.989	0.985	0.137	0.291	0.000	0.234
OLMo-2-1B	0.000	0.100	0.750	0.910	0.844	0.828	0.130	0.005	0.074

Table 9: Diagnostics for the seven heads found on FLORES, as $C \rightarrow W$ on 2,500 FLORES prompts except for the statistics in (a) and the Swapped-in language columns in (b) (columns defined in the text). –: not run. Zero and Prompt mean may differ slightly from Table 1 because these runs use per-language batches. With any of three same-layer control heads swapped instead, no LCB reply is in the swapped-in language.

Base model	Our instruct head	Directions topped	Mean-score rank
Qwen3-1.7B	L18H12	20/20	1/448
Llama-3.2-1B	L8H25	7/20	1/512
Gemma-3-1B	L11H3	0/20	3/104

Table 10: Instruction-tuned heads under the Translation Heads ranking on released base-model scores. In Llama-3.2-1B, L9H8 ties L8H25 in the number of directions topped.